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Neural Network

Objective

The aim of this analysis is to predict whether an observation corresponds to a high-consumption event (`high_consumption`), a binary outcome taking values 0 or 1. Unlike the Poisson GLM, which models counts , this setting is a classification problem.

A neural network is used to capture potentially non-linear relationships between consumption and predictors such as weather, building characteristics, and household composition. The focus is on predictive performance, with probabilities converted into class predictions and evaluated on test data using accuracy, sensitivity, and specificity.

Data preparation

The dataset consists of 99,838 observations and includes variables related to weather conditions, building characteristics, and household composition . The response variable, `high_consumption`, is binary and approximately balanced between the two classes.

To ensure valid model evaluation, the data is split into training (80%) and test (20%) sets using a stratified sampling approach, preserving the class distribution in both subsets. This avoids bias in performance estimates.

Categorical variables are converted into dummy variables using one-hot encoding, and all predictors are scaled based on the training data. The same transformation is then applied to the test set, ensuring consistency and preventing data leakage.

Model setup

A feedforward neural network was fitted to classify whether an observation corresponds to a high-consumption event. The model uses 8 input variables, one hidden layer with 3 neurons, and 1 output node, corresponding to an **8-3-1 network** .

In compact form, the model can be written as:

$$\hat{y} * i = f\left(\sum *j = 1^3 w_j^{(2)}, g\left(\sum_{k=1}^{8} w_{jk}^{(1)} x_{ik} + b_j^{(1)}\right) + b^{(2)}\right)$$

where x_{ik} denotes the input predictors, $g(\cdot)$ is the hidden-layer activation function, and $f(\cdot)$ produces the predicted probability of high consumption.

The model was estimated using the `nnet` package with a weight decay parameter (0.001) to regularise the network and reduce overfitting, and a maximum of 300 iterations to ensure convergence . All categorical predictors were dummy-encoded and inputs were scaled prior to training.

Model Results

Model performance is evaluated on the held-out test set using standard classification metrics. The neural network achieves an overall accuracy of 0.770, indicating that approximately 77% of observations are correctly classified.

Table 1: Neural Network Performance Metrics

Model	Accuracy	Sensitivity	Specificity
Neural Network	0.77	0.803	0.737

Sensitivity (true positive rate) is 0.803, showing that the model performs well in identifying high-consumption events. Specificity (true negative rate) is slightly lower at 0.737, indicating somewhat weaker performance in correctly classifying low-consumption cases.

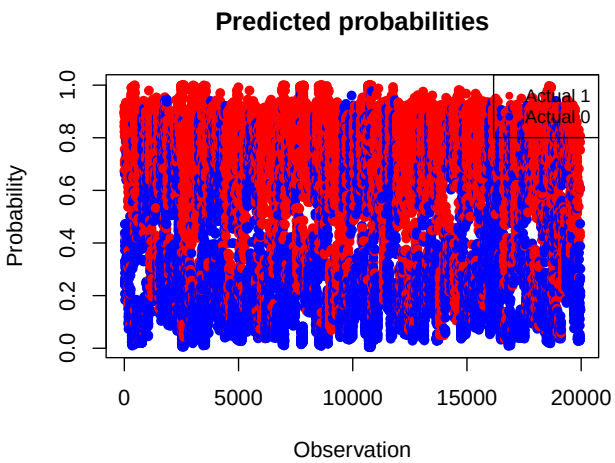
Overall, the results suggest that the model has good predictive performance, with a balanced ability to detect both classes, although it is slightly more effective at identifying high-consumption observations.

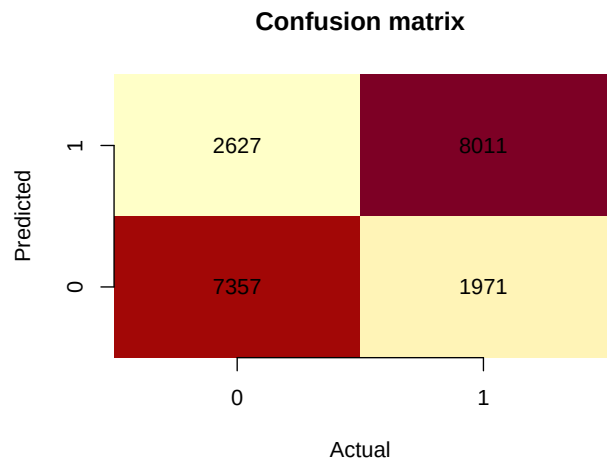
Visual Evidence

Visual Evidence

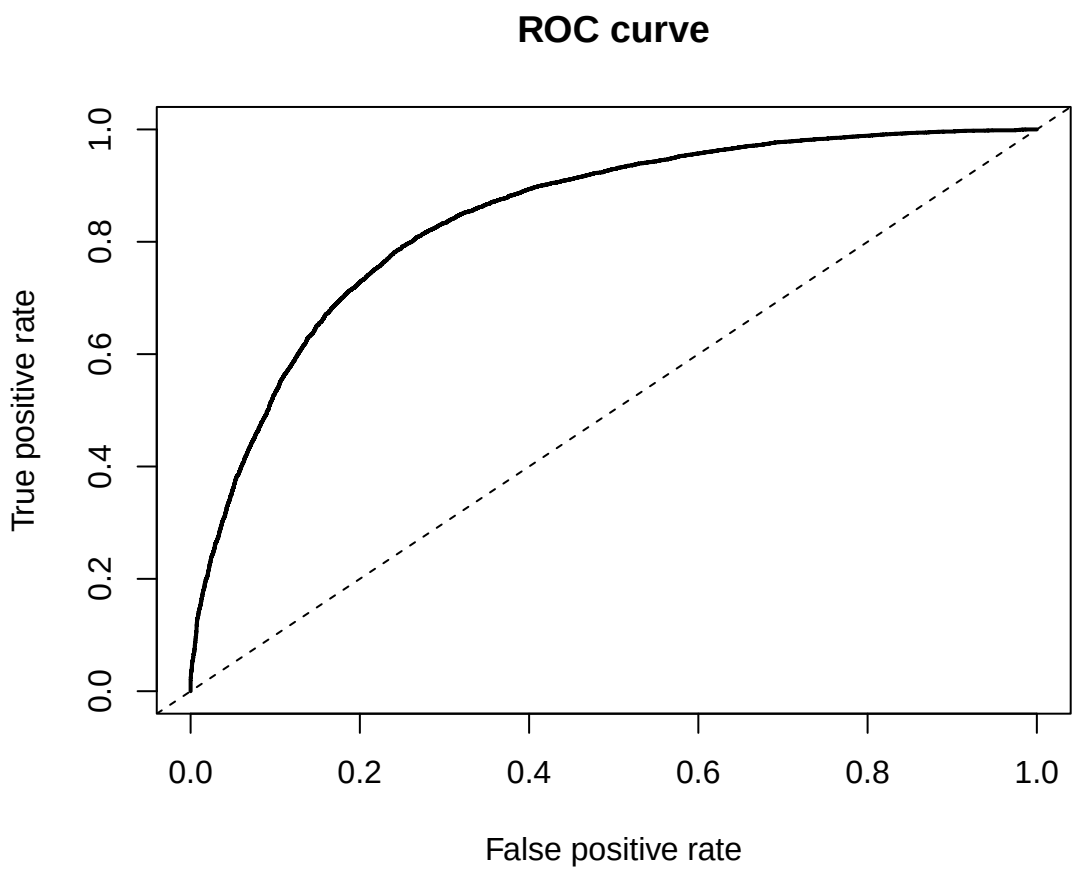
The *predicted probability plot* shows how the neural network separates the two classes. Observations belonging to the high-consumption class (red) tend to have higher predicted probabilities, while low-consumption observations (blue) are more concentrated at lower values. However, there is substantial overlap between the two groups, particularly in the mid-range of probabilities. This indicates that, although the model captures meaningful patterns, it cannot perfectly distinguish between the two classes, which is consistent with the observed classification errors.

The *confusion matrix* provides a direct summary of classification performance. The majority of observations lie on the diagonal, indicating correct predictions for both high- and low-consumption cases. The model correctly identifies a large number of high-consumption events, as reflected in the high true positive count, while also achieving a substantial number of correct low-consumption classifications. Misclassifications occur in both directions, with some low-consumption observations predicted as high and vice versa, confirming that class separation is not complete.





Finally, the *ROC curve* summarises model performance across all possible classification thresholds. The curve lies clearly above the diagonal reference line, indicating that the model performs substantially better than random classification. The steep initial increase shows that high true positive rates can be achieved with relatively low false positive rates, while the gradual flattening at higher values reflects diminishing gains. Overall, the shape of the curve suggests good discriminative ability, supporting the results obtained from the confusion matrix and accuracy metrics.



Model Diagnostics

Key Findings

Conclusion